

David Frieв

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About Me

I write in C++ and Go, 2nd year student of Applied Mathematics and Computer Science at Kosygin State University of Russia. I work on my own projects: most of my time goes to a game engine in C++ and its asset pipeline in Rust, I also write a microservice search engine in Go. I want to study system architecture and grow in the development of high-load services.

Projects

Search Engine

Go, Docker

A microservice search system. Web crawler with Redis queue and robots.txt support, BM25 indexing via Meilisearch, REST API in Go, frontend in Nuxt. Infrastructure is deployed through Docker Compose. Was deployed on Yandex Cloud at search.koftamainee.ru; currently inactive due to trial credits depletion.

In development: a metrics service in Rust for collecting page-quality signals and PageRank on the link graph for result re-ranking.

Mantle

C++, Vulkan

A modular game engine. Vulkan renderer with Frame Graph for automatic GPU synchronization, virtual memory system with custom allocators (arena, tslf), integration with flecs for ECS architecture and Jolt Physics for physics simulation, custom scripts for code generation and formatting, CI/CD with auto-builds for Linux and Windows.

In development: extracting a stable Engine API for game code integration, writing high-level abstractions for asset and scene loading, and customizing rendering from game code.

Mint / Mintload

Rust, C

Asset pipeline for Mantle (canonically Mantle INput Tool): converts glTF/GLB to binary formats suitable for direct GPU upload, scene system via text .mscn files and conversion to binary .mscb format, project management system and incremental builds. Mintload — a small C library for fast file loading into the engine via mmap, with a C++ wrapper.

In development: shader compiler (.mshd), hot-reload support, release archive format (.mpak).

Other

[Chalkboard](#) — a C++20 library for rendering LaTeX formulas with hot-reload.

[Danmaku](#) — a game engine in pure C with Lua scripting (SDL3). Currently on hold, but planning to rewrite rendering in Vulkan and finish the project.

Skills

Languages: C/C++, Go, Rust, C, Lua

Technologies: Vulkan, Docker, Linux

Education

Kosygin State University of Russia, Moscow

Applied Mathematics and Computer Science, 2nd year (2024–2028)